

Board Specs

1. OMAP3530 Processor
 - 600MHz ARM Cortex-A8 (>1200 DMIPS)
 - Over 1 billion dhrystones per second with NEON and VFP extensions
 - 2D/3D graphics accelerator
 - C64x+ DSP and video acceleration
2. 256MB DDR@166MHz
3. 256MB NAND
4. High-speed USB 2.0 OTG
5. High-speed USB HOST
6. S-Video (TV out)
7. DVI-D
8. Stereo Audio Out & In
9. 8-bit SD/MMC
10. JTAG
11. UART
12. Expansion connector to provide extended connectivity with standard connections like:
 - FC, MMC, SPI, GPIO
 - LCD expansion—the new LCD interface can be used with an adaptor board to connect to displays, including RGB LCD panels and LVDS panels
 - PWM—the expansion connector now provides three pulse-width modulation (PWM) signals for motor driver functions or PWM signalling typically used in robotics

(1400mAh) and about 4 hours of video (1400mAh)."

Moreover, the Beagle Board is also provided with a S-Video and a DVI-D port. These can be used for connecting to a TV and a monitor, respectively. The UART is a serial port and it helps a developer in kernel-level debugging. The

The board is designed to use standard (PC) peripherals like a DVI, keyboard, mouse, Ethernet, Bluetooth, Wi-Fi and hard disk (over USB). You can upgrade, change or expand these peripherals based on your need

JTAG cable that is provided can be used for JTAG-based simulator and low-level debugging. The device also has an 8-bit SD/MMC, which can be used as a hard disk. The PWM expansion connector provides three pulse-width modulation (PWM) signals for motor driver functions or PWM signalling typically used in robotics.

Now, what about the scope for expandability of the board? The board is designed to use standard (PC) peripherals like a DVI, keyboard, mouse, Ethernet, Bluetooth, Wi-Fi and hard disk (over USB). You can upgrade, change or expand these peripherals based on your need. The board also provides an expansion header that provides OMAP3530 pin-outs to interface with any external device over SPI, MMC, FC and GPIO interfaces. A few examples are:

1. Beagle is used by an open community member to build an open software-defined radio—www.opensdr.com

2. The expansion header can be used to connect this to an analogue-to-digital converter and process the live samples on DSP and use ARM for the

host interface

3. TI DLP Pico projector can be interfaced through the DVI interface on Beagle Board and the framebuffer content can be projected (dcl1.digikey.com/us/mkt/pico.html)

4. With Angstrom distribution ported, the board can be used as a regular desktop

5. Android is supported by Beagle-Board.org community

6. u-buntu (the Canonical port for ARM Cortex A8) will bring in thousands of apps on Beagle Board

The bottom line is that, as a developer, you can use it for Web services, 3D gaming, the Linux kernel and drivers, boot loaders and firmware, portable media and infotainment applications, the UI framework, high-end signal processing with generic DSP, and OLPC applications like Sugar—in short, the possibilities are endless; you'll only have to let your imagination run wild. Just to give you an idea, you might want to check out all the current projects that are under way at www.beagleboard.com/projects. So what are you waiting for? Take a bite at this 'Open Platform'.

Syed notes, "While there are numerous theoretical concepts available today, these could be executed with 'Open Platforms,' which is a new trend in the IT industry where products are defined, designed and developed by consumers across the world. Ideas are just like raw materials. They need enormous energy, effort and tools to convert them into end products. Collaboration in such efforts results in better execution." ●

The article is reproduced from June 2009 issue of LINUX For You (LFY) magazine; both the authors are part of LFY Bureau

References and documentation

- Beagle Board community portal: beagleboard.org
- Technical reference manual for OMAP3530 processor: focus.ti.com/general/docs/gencontent.tsp?contentid=36915
- DSP LINK (for ARM-DSP communication): www.a-ti.com/downloads/sds_support/targetcontent/link/link_1_50/index.html
- Validation software, Linux device drivers, latest kernel from GIT: code.google.com/p/beagleboard/wiki/BeagleboardRevCValidation
- Board schematics: beagleboard.org/static/BBSRM_latest.pdf
- DSP Compiler: www.a-ti.com/downloads/sds_support/targetcontent/LinuxDspTools/download.html
- ARM Compiler: www.codesourcery.com/gnu_toolchains/arm/portal/release313
- More links and documentation: elinux.org/BeagleBoard & code.google.com/p/beagleboard
- Discussion list: discussion at beagleboard.org
- IRC: #beagle on irc.freenode.net